

Is Metacognition a Better predictor than IQ for Academic Achievement?

A Methodological and Psychometric Critique of
Gomes, Golino & Menezes (2014), *Psychology*, 5, 1095–1110

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Abstract

In their disturbingly widely cited paper, Gomes, Golino and Menezes (2014) report that general metacognitive ability explains 20.43% of general academic achievement against just 7.45% for fluid intelligence, and that specific metacognitive ability explains 42.12% of specific academic achievement against 1.12% for fluid intelligence, with advertised ratios of “nine times” and “15:1 at worst” in favour of metacognition. My article shows that these results are manufactured by several complementary defects, which should not have escaped the eyes of an unbiased and diligent researcher, much less those of 3 PhD’s. First, the scoring rule of the mathematics appraisal test penalizes accurate self-knowledge; for example, a student with zero metacognition who simply declares certainty of success on every item is guaranteed a higher “metacognition” score than a perfectly self-aware student, at every accuracy level below 100% (Proposition 1); the score is, in significant substance, an arithmetic achievement measure. Second, the criterion for specific achievement is a parallel arithmetic test, so the headline association sits comfortably below the ceiling implied by shared accuracy alone. Third, the intelligence covariate, an obscure, short, and fluid-only battery ($\alpha = .75$) in a single-school, range-restricted sample, is reproduced to three decimal places by applying standard range-restriction and unreliability corrections to the conventional population correlation of $\rho = .50$ (cited by the authors as well as the standard value of reference), so the data contain no evidence against the primacy of intelligence. Model-side choices (factor scores treated as error-free, modification-index respecification, a directed path from metacognition to intelligence (even though, in my opinion, it should be the other way around), and effect ratios computed between the extremes of different confidence intervals) further illustrate the poor quality of the study. All computations are stated explicitly and all empirical data is cited to page.

Note on the qualifications of the author. I ought to make the reader aware that I am an undergraduate law student and not a formally qualified expert in this subject matter. My critique should be treated with increased scrutiny. Nonetheless I made every effort to ensure the quality of this work is up to standard.

1 The paper and its claims

Gomes, Golino and Menezes [1] administered a fluid-intelligence battery, two metacognitive tests, and an arithmetic test (then used to quantify specific academic achievement) to 684 students (grades 6–12) of a single private school in Belo Horizonte, Brazil (p. 1098), and combined these with annual grades in Mathematics, Portuguese, Geography, and History in a structural

equation model. Their central estimates (p. 1103), reproduced in Table 1, ground three claims: that metacognition has incremental validity over intelligence; that in “the best of the scenarios” general metacognitive ability explains general achievement “approximately nine times more” than fluid intelligence (p. 1104); and that specific metacognitive ability outpredicts intelligence “at worst in 15:1” and prior mathematical knowledge “at worst in 18:1” for specific achievement (p. 1104).

The Critique is organized around three portant elements of the design: the scoring rule of the metacognition measure (Section 2), the relation between predictor and criterion (Section 3), and the construction of the intelligence covariate (Section 4); Section 5 collects the model-side choices and Section 6 states what a probative design would require.

2 Scoring rule penalizes accurate self-knowledge

2.1 The rubric

The *Appraisals Ability on Mathematics Expressions* test (AAME) asks students to solve 18 arithmetic expressions and, after each, to judge their probability of success on a four-point scale (p. 1099). The published scoring rule (p. 1099) is summarized in Table 2. Its defining feature: a student who is sure of having failed receives 0 points *even when that judgment is correct*. The authors justify the design intention with the following statements: “students should be minimally able to solve the item in order to perform the evaluative process” and “This strategy can be also justified by the avoidance of measuring other construct than self-appraisal, like the feeling-of-not-knowing [...] or the feeling-of-knowing” (p. 1099), which means that the contamination of the metacognition score by task performance is not an accident of analysis, but rather a conscious choice.

Metacognitive monitoring, as the field defines it, is the correspondence between confidence and performance; a valid monitoring score must reward correct self-assessments of failure as much as correct self-assessments of success. Table 2 does the opposite, and the consequences can be estimated.

Table 1: Variance-explained estimates reported by Gomes et al. (2014, p. 1103), with their 90% bootstrap confidence intervals. GMA = General Metacognitive Ability; SMA = Specific Metacognitive Ability; GAA/SAA = General/Specific Academic Achievement; Gf = fluid intelligence; MI = Monitoring Indicator (reading test); AI1/AI2 = appraisal indicators built from the easy/difficult arithmetic items.

Path	Interpretation	Variance explained	90% CI
GMA → GAA	metacognition → school grades	20.43%	[12.60, 29.38]
Gf → GAA	intelligence → school grades	7.45%	[3.31, 12.46]
SMA → SAA	metacognition → arithmetic	42.12%	[36.00, 48.44]
MATH → SAA	prior math grade → arithmetic	3.72%	[1.99, 5.86]
Gf → SAA	intelligence → arithmetic	1.12%	[0.32, 2.34]
GMA → MI		66.10%	[49.00, 99.06]
GMA → Gf	metacognition “explains” intelligence	12.85%	[6.05, 21.25]
SMA → AI1	easy-item indicator	90.25%	[81.36, 100]
SMA → AI2	difficult-item indicator	28.09%	[22.66, 33.64]

Table 2: The AAME scoring matrix implied by the verbal rubric of Gomes et al. (2014, p. 1099).

Confidence judgment	Item solved correctly	Item solved incorrectly
“Sure I failed”	0	0
“Not sure; think I failed”	2	3
“Not sure; think I succeeded”	3	2
“Sure I succeeded”	4	1

2.2 Expected-score analysis

Let p denote a student’s probability of solving an arithmetic item correctly, constant across the 18 items for simplicity’s sake. Fix a *reporting policy*, i.e., a rule mapping the student’s (possibly absent) self-knowledge to a confidence response, and compute the expected total score $E[S]$ under Table 2.

- **Perfect self-knowledge, honest reporting.** The student is always certain and always right about himself: “sure I succeeded” on correct items (4 points), “sure I failed” on incorrect items (0 points):

$$E[S_{\text{cal}}] = 18[4p + 0(1 - p)] = 72p. \quad (1)$$

- **Zero self-knowledge, blind overconfidence.** The student selects “sure I succeeded” on every item, ignoring all internal evidence (4 points if correct, 1 if not):

$$E[S_{\text{over}}] = 18[4p + 1(1 - p)] = 18 + 54p. \quad (2)$$

- **Zero self-knowledge, random confidence.** The student chooses randomly among the four responses. Correct items then earn $(0 + 2 + 3 + 4)/4 = 2.25$ points on average and incorrect items $(0 + 1 + 3 + 2)/4 = 1.5$ points:

$$E[S_{\text{rand}}] = 18[2.25p + 1.5(1 - p)] = 27 + 13.5p. \quad (3)$$

- **Perfect self-knowledge, optimized reporting.** A self-aware student who games the rubric, i.e. he answers “sure I succeeded” when he knows he is right (4) and “not sure; think I failed” when he knows he is wrong (3, the maximum available for an incorrect item):

$$E[S_{\text{opt}}] = 18[4p + 3(1 - p)] = 54 + 18p. \quad (4)$$

Proposition 1 (Blind overconfidence outperforms honest calibration). *For every $p < 1$, $E[S_{\text{over}}] - E[S_{\text{cal}}] = 18(1 - p) > 0$: a student with no metacognition at all, i.e. one who always claims certainty of success, always obtains a higher “metacognition” score than a student with perfect, honestly reported self-knowledge. The two coincide only at $p = 1$ i.e. when they both answer every arithmetic question correctly.*

Proposition 2 (Random confidence beats calibration below 46% accuracy). *Setting (1) equal to (3) gives $72p = 27 + 13.5p$, i.e. $p = 27/58.5 \approx .462$. For any student correctly solving fewer than 46% of the items, answering the confidence question randomly yields a higher expected score than perfect self-knowledge.*

Proposition 3 (Honesty is penalized even given perfect self-knowledge). *Comparing (4) with (1): $E[S_{\text{opt}}] - E[S_{\text{cal}}] = (54 + 18p) - 72p = 54(1 - p) \geq 0$. The score-maximizing policy for a*

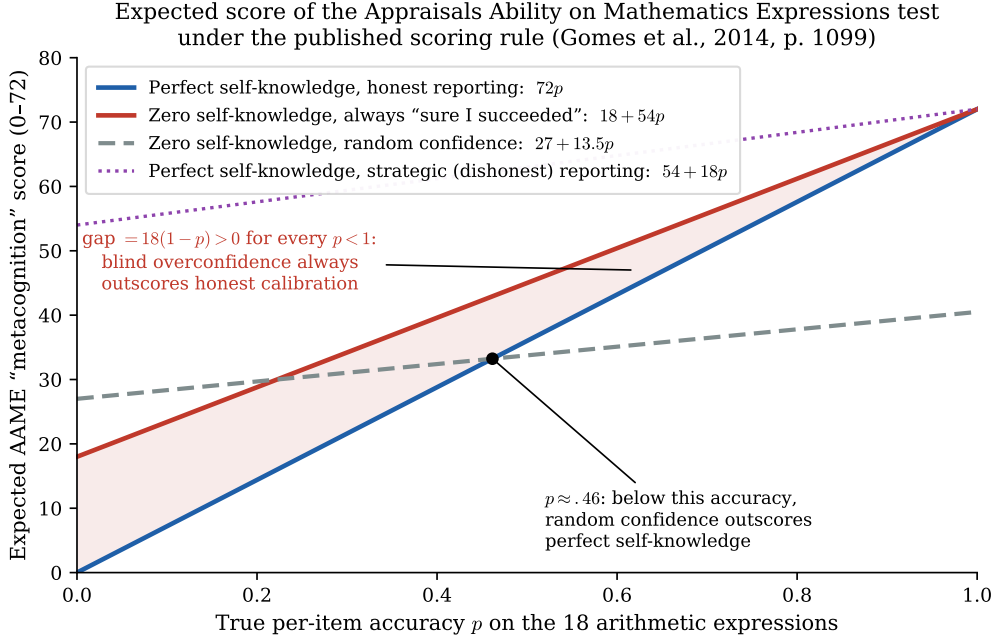


Figure 1: Expected AAME score as a function of true item accuracy p under four reporting policies (equations (1)–(4)). The shaded band is the dominance gap of Proposition 1. Figure computed by the author from the scoring rule on p. 1099 of [1].

perfectly self-aware student is to lie about his certainty of failure (avoiding the 0-scored “sure I failed” response). A test on which the optimal strategy for the best possible metacognizer is to lie about his metacognition is not properly measuring metacognition.

Figure 1 plots (1)–(4). Two further readings of the matter. First, under every policy the expected score is increasing and linear in p , with accuracy slopes of 72, 54, 18 and 13.5 points per unit of p ; whatever a student’s confidence policy, his score rises mechanically with his arithmetic ability, so between-student variance in the AAME total is dominated by achievement variance (emphasis on “variance”). Second, at any fixed p the vertical spread between “perfect” and “zero” metacognition policies is small and *non-monotone in metacognitive quality* (the honest-calibrated line lies *below* two zero-metacognition lines over most of the range). The instrument is, in mathematical substance, an arithmetic achievement test with a confidence-flavoured perturbation.

The paper’s own factor results corroborate this diagnosis: latent Specific Metacognitive Ability explains 90.25% of the indicator built from the *easy* arithmetic items but only 28.09% of the indicator built from the difficult ones (Table 1; Figure 2b). Easy items are where confidence responses sit at ceiling and residual score variance reduces to accuracy variance; the “metacognition” factor lives where metacognition supposedly varies least.

3 The criterion is almost the same test as the predictor

What does the contaminated score predict? “Specific Academic Achievement” is measured by the *Arithmetic Expressions Test*: 18 arithmetic expressions scored pass/fail, with Cronbach’s $\alpha = .88$ (p. 1100); imagine my shock upon finding out about this. The appraisal test (the one used to estimate Specific Metacognitive Ability) itself consists in the solving of 18 arithmetic expressions (p. 1099), with $\alpha = .86$ (p. 1100). The paper never states that the two item sets

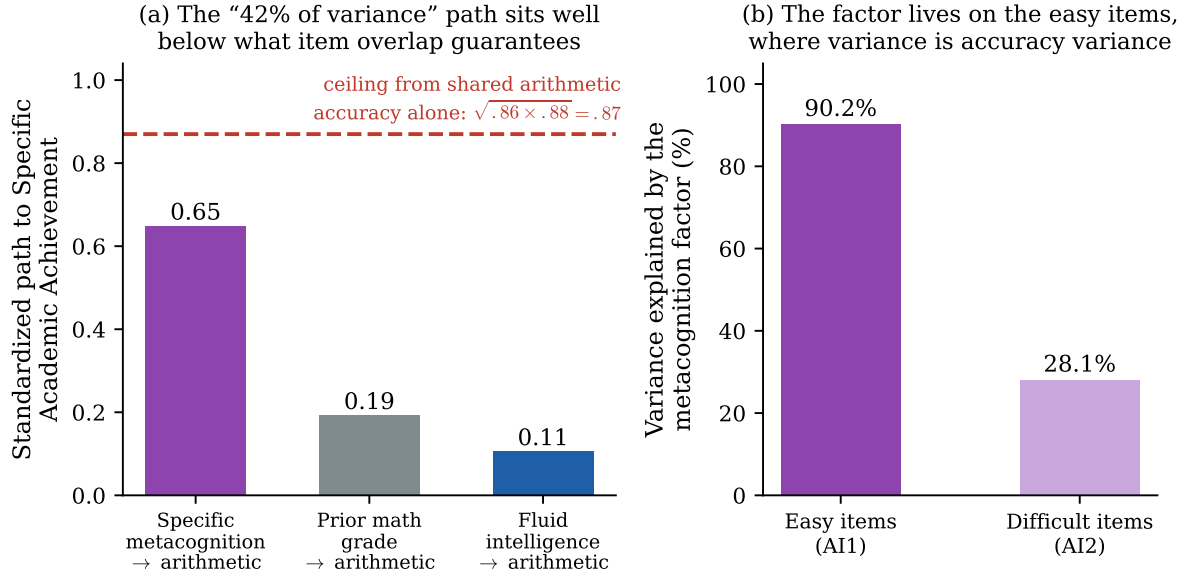


Figure 2: (a) The specific-achievement path against the attenuation ceiling of equation (5): the celebrated association is most likely explicable by shared arithmetic accuracy. (b) Variance in the two appraisal indicators explained by the metacognition factor (p. 1103): the factor is anchored in the easy items, where score variance is accuracy variance.

are distinct, and the appraisal instructions (“After solving the item, the respondents must evaluate their probability of success,” p. 1099) describe a single solve-then-judge procedure. On my most natural reading the criterion scores are the solving outcomes in the predictor’s own administration, a part-whole relation; on the most charitable reading they are parallel forms of the same type of test taken in the same session.

Either way the association is guaranteed in advance. Both totals are functions of the same underlying accuracy p , so their correlation is limited only by measurement error. By the classical Spearman attenuation identity [2], the maximum observable correlation between two fallible measures of a common quantity is

$$r_{\max} = \sqrt{r_{xx} r_{yy}} = \sqrt{.86 \times .88} \approx .870. \quad (5)$$

The paper’s standardized path from Specific Metacognitive Ability to Specific Academic Achievement is $\sqrt{.4212} \approx .649$, which is comfortably inside the range that shared arithmetic accuracy alone produces, with no metacognitive contribution required at all (Figure 2a).¹ The headline claims about predictive power, i.e. metacognition 42.12%, intelligence 1.12%, prior mathematics grades 3.72% (p. 1103), therefore translates as: “performance on 18 arithmetic expressions predicts performance on similar 18 arithmetic expressions better than an abstract-reasoning test or last year’s grades do”.

The “general” side has the same issue in milder form. The Reading Monitoring Test (find nine planted contradictions in a text (p. 1098)) is an error-detection task, classically confounded with reading comprehension and verbal ability: a reader who misses a contradiction may simply not have understood the passage [5, 6]. The authors themselves mention this critique of the Markman paradigm on p. 1099 before adopting the paradigm anyway, and the test’s reliability

¹The .649 is a path coefficient estimated with fluid intelligence and prior mathematics grades also in the equation albeit flawingly so as I’ll explain later; the corresponding zero-order association is, if anything, larger, which only strengthens the point.

is a modest $\alpha = .63$ (p. 1100). The General Metacognitive Ability latent, assembled from a verbal-comprehension-loaded reading task and an arithmetic-performance-loaded appraisal task, is operationally a verbal-plus-numerical scholastic composite. That such a composite predicts school grades in Portuguese, Mathematics, History and Geography isn't unexpected, and does not prove a strong correlation between metacognition and academic achievement.

4 Dubious methodology for the estimation of fluid intelligence

Against this scholastic composite, “intelligence” was estimated with a single reduced fluid-intelligence test (the “CTIF” test kit) with which I was not familiar and rightly so, given that it is indeed an obscure test: the only references I could find about it were in articles produced by co-authors of the presently critiqued metacognition study; most concerningly, the test itself was co-developed by one of the co-authors [14]. The reduced test consists of 27 items, Cronbach's $\alpha = .75$ (falling to .69 in the authors' own re-analysis for the reduced version), with eight items loading below .40 and one at .05 (p. 1100). Most notably, this test lacked any item assessing crystallized intelligence, or verbal factor, and no general factor was extracted from multiple broad abilities, despite a criterion (grades in language- and knowledge-heavy subjects) that loads heavily on crystallized ability. Having excluded verbal ability from the intelligence side of the model, the design then lets the reading test carry that same variance under the label “metacognition”. The single-school sample (p. 1098; acknowledged in pp. 1106–1107) compounds the handicap: school selection restricts the spread of intelligence, and restriction of a predictor's spread obviously shrinks its correlations.

The paper's own results should be symptomatic of the poor quality and artificial direction of this test: Fluid intelligence explains 7.45% of general achievement (p. 1103), yet the introduction cites the established finding that intelligence explains 25–50% of achievement variance (p. 1096), and Deary and colleagues' five-year study of some 70,000 English schoolchildren, which was cited on that same page found a latent correlation of about .81 between intelligence at age 11 and national examination results at 16, i.e. 66% of variance at the latent level [4]. When a benchmark underperforms the literature by a factor of three to nine, authors should assume that the defect likely lies in the measurement. This can be made quantitative. The reported 7.45% corresponds to a standardized coefficient of $\sqrt{.0745} \approx .273$. Start from the conventional population correlation $\rho = .50$ (which the authors themselves cite as the standard) and apply the two mentioned artifacts in sequence.

1: range restriction. Thorndike's Case II formula [3] corrects for restriction of range in such analysis inflating the observed value in the restricted sample back to the true population value, and though I'm no statistician, this seems to me like the appropriate instrument (admittedly found and applied through the power of google). I inverted the original formula, so starting from a presumed unrestricted correlation instead of an observed correlation. Writing $u = \sigma_{\text{restricted}}/\sigma_{\text{population}}$,

$$r_{\text{restr}} = \frac{u \rho}{\sqrt{u^2 \rho^2 + 1 - \rho^2}} = \frac{0.60 \times 0.50}{\sqrt{0.36 \times 0.25 + 0.75}} = \frac{0.30}{\sqrt{0.84}} \approx 0.327, \quad (6)$$

taking $u = 0.60$ as plausible degree of selection for a private school (an assumption, this isn't an estimate from their data, though one must consider the strong correlation between socioeconomic factors of parents and IQ of offspring, as well as possible entrance tests; such correlation has never, to my knowledge, been established for metacognition).

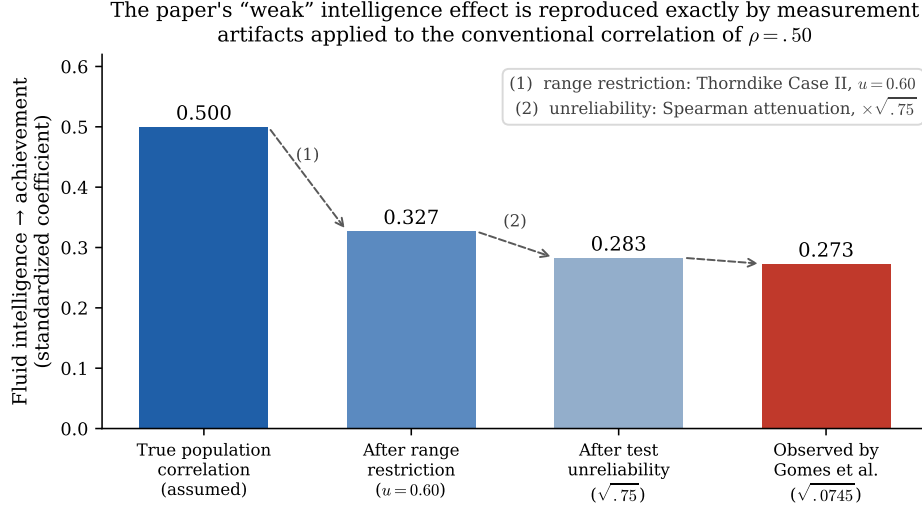


Figure 3: Reconstruction of the paper’s fluid-intelligence coefficient from probable artifacts: the conventional $\rho = .50$ (which they themselves cite as the standard), passed through Thorndike’s Case II range-restriction formula (6) with an assumed selection ratio $u = 0.60$ and the Spearman attenuation formula (7) with the reported reliability $\alpha = .75$, lands on the observed value to within 0.01.

2: unreliability. Attenuation for predictor unreliability multiplies the correlation by the square root of the measure’s reliability:

$$r_{\text{obs}} = r_{\text{restr}} \times \sqrt{r_{xx}} = 0.327 \times \sqrt{.75} \approx 0.283 \quad (\text{or } 0.327 \times \sqrt{.69} \approx 0.272). \quad (7)$$

The observed .273 is recovered almost exactly (Figure 3). The conclusion is not that $\rho = .50$ is so proven, since u was assumed, but rather that the paper’s data are probably consistent with the standard intelligence–achievement correlation after proper adjustment and therefore contain little evidence against the primacy of intelligence. In other words, they show what $\rho \approx .50$ looks like after passing through a short, noisy, fluid-only, and dubious test in a range-restricted sample (It’s important to stress again that $\alpha = .75$ is their best case scenario for reliability, as the paper itself reports $\alpha = .69$ for this reduced version of their test) in this case. The same logic is the covariate-side threat: a control variable measured with error cannot absorb its own variance, and the unabsorbed remainder strengthens whatever correlated predictor stays in the model [7], in this case, the scholastic composite which the authors label as metacognition.

5 Model choices that allocate shared variance to metacognition

Four analytic decisions then tilt whatever variance remains in favor of metacognition, mostly at the expense of fluid intelligence.

Factor scores treated as error-free. The authors claim factor scores “take into account only the true score and eliminate both the error and specific variance” (p. 1101). This is incorrect as factor scores are fallible estimates subject to factor indeterminacy [8, 9]. This means that entering them as single observed variables, as opposed to latent variables, completely loses the disattenuation that a full latent-variable specification would have provided, i.e. the least reliable measure in the model, which is the intelligence, takes the hit. battery.

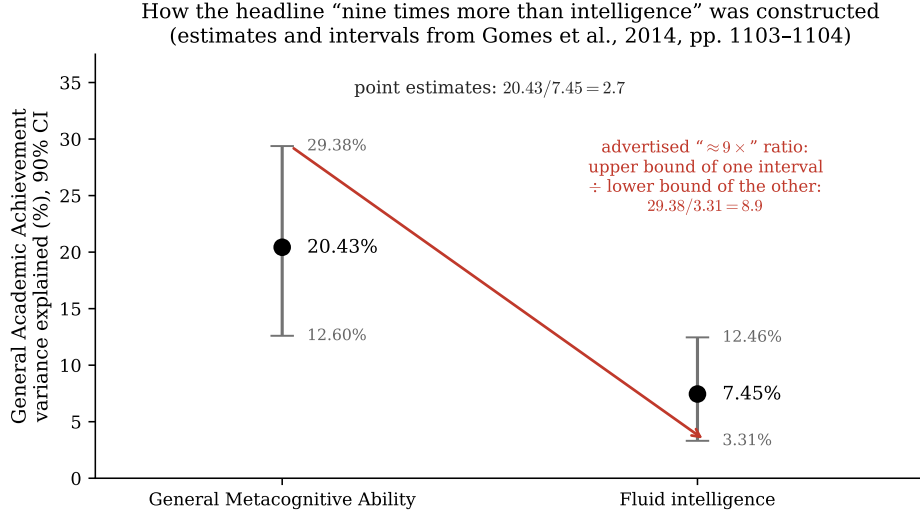


Figure 4: The construction of the headline ratio. Point estimates and 90% intervals from p. 1103 of [1]; the “ $\approx 9\times$ ” claim (p. 1104) divides the upper bound of one interval by the lower bound of the other.

Modification-index respecification. The final model was reached by adding paths suggested by modification indices (a Mathematics–Portuguese covariance; direct paths from fluid intelligence to Mathematics and History), improving fit from $\chi^2 = 103.31$ ($df = 22$) to $\chi^2 = 51.18$ ($df = 19$), CFI = 1.00 (p. 1103). Post-hoc respecification of this kind capitalizes on sample-specific chance, and a near-saturated model with CFI = 1.00 cannot then be offered as confirmation.

A directed path from metacognition to intelligence. The model routes $GMA \rightarrow Gf$, with metacognition “explaining” 12.85% of intelligence (p. 1103), a pure specification choice in cross-sectional data that credits the shared variance of the two constructs to metacognition by construction. The discussion states the mechanism with disarming obliviousness: when the metacognition trait enters, “it extracts considerably the intelligence’s variance” (p. 1105). That sentence describes how variance was allocated under an arbitrary model, not a genuine finding.

Ratios computed between the extremes of different intervals. The advertised “approximately nine times” (p. 1104) is obtained by dividing the *upper* bound of metacognition’s 90% interval by the *lower* bound of intelligence’s: $29.38/3.31 = 8.9$. The point estimates give $20.43/7.45 = 2.7$ (Figure 4). The “15:1” and “18:1” figures for specific achievement are the same construction ($36.00/2.34 = 15.4$; $36.00/1.99 = 18.1$). Comparing the top of one confidence interval with the bottom of another is not a legitimate effect-size comparison under any convention; and I can’t think of any good reasons why anyone would choose to present the data in this manner.

6 Conclusion

Set beside the think-aloud literature of the Veenman group [10, 11] and the accuracy-measure tradition, this paper is an exemplary specimen of a recurring recipe for “metacognition beats intelligence” results: (i) the metacognition measure is taken on or immediately beside the criterion task while intelligence is measured remotely; (ii) the metacognition score is contaminated with

raw performance, and here the contamination is written into the rubric itself (Section 2); (iii) the intelligence covariate is short, unreliable, fluid-only, and administered in a range-restricted sample (Section 4); (iv) the model is tuned after the fact and the directional choices favour metacognition (Section 5); and (v) personality, motivation and prior achievement are absent or accounted for (note that even prior mathematics grades were permitted only 3.72% here, because the model routed the shared variance through metacognition first).

The paper is also instructive for what it fixed. It repaired two weaknesses it identified in the earlier think-aloud studies, i.e., judge-based scoring, and small samples (p. 1097), but failed on construct contamination and covariate quality, and it thereby produced the most inflated metacognition-to-intelligence ratio in the literature (as far as I am aware). A good design would need to satisfy the following three conditions (at the very least): a criterion measured independently of the metacognition assessment occasion and item pool; an intelligence covariate measured at full strength (multiple broad factors, adequate reliability, unrestricted or restriction-corrected sample, preferably using a recognized IQ test such as the WISC or the WAIS); and a metacognition score mathematically decoupled from raw academic performance. As far as I am aware, no study claiming metacognitive superiority over intelligence has yet met all three at once. Where all three are approximated, the unique predictive contribution of metacognition settles into the 1–3% of criterion variance in my private assessment from the meta-analytic evidence [12], i.e., an order of magnitude below the ratios advertised here, but I should again stress how mine, is just a very unreliable estimate and I would not stake much on it. What I can say with much greater certainty is that the currently available evidence suggests that the predictive power of metacognition is extremely weak, and certainly much weaker than that of intelligence.

Note on sources and computation. All empirical quantities attributed to [1] are cited to page above. All figures were computed by me from the equations and estimates stated in the text; no data beyond the published estimates were used. The selection ratio $u = 0.60$ in Section 4 is explicitly an assumption; every other input is taken from the paper.

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